

EDUCATION

University of Delaware

B.S. Computer Engineering; Minor: Computer Science & Cybersecurity

Newark, DE

Feb 2021 - May 2025

- **GPA:** 3.44/4.0; **Dean's List:** Spring 2021, Fall 2022, Fall 2024, Spring 2025

EXPERIENCE

Hologic, Inc

Glasgow, DE

Software Engineer Intern

June. 2026 - Sep. 2026 (Expected)

- Developed a C#/WPF desktop application to decode and process Hologic's Mammography System gantry NVM files, converting raw hexadecimal control data across 19 hardware nodes into readable configuration.
- Implemented node identification, checksum/range validation, and NVM file regeneration to verify configuration integrity and support reliable gantry updates.
- Collaborated with software, systems, and hardware engineers in an Agile/Scrum development environment to develop and validate node communication, startup/POST sequences, and system-level behavior.
- Developed software test cases to verify NVM decoding accuracy and troubleshoot defects using Git and Perforce-based version control, ensuring reliability of gantry configuration data prior to deployment.

DEVCOM Research Intern + Senior Design

Newark, DE

Compact and Fast-scanning passive mmWAVE imager

June. 2024 - May. 2025

- Designed a compact, passive imaging device that detects mmWave radiation for security screenings.
- Developed embedded control software in Python for a Raspberry Pi 5, incorporating a specialized Farran radiometer with motorized linear movement and ADC-based data acquisition.
- Built a Python GUI and backend workflow for scan control, heatmap generation, and AI-assisted scan feedback, using FastAPI POST endpoints for real-time heuristic threat analysis of scans.
- Engineered a custom 4-layer PCB to process and low-amplitude mmWave signals, improving signal integrity strength.
- Advanced a senior design project at the University of Delaware, preparing an academic poster and research paper.

University of Delaware

Newark, DE

Researcher - Trustworthy Artificial Intelligence Development

Feb. 2024 - May 2024

- Created models to categorize underwater terrain and sonar reflections using Explainable AI (XAI).
- Conducted experiments using PyTorch library to evaluate model applicability across different datasets.

Department of Mathematical Sciences and Office of Academic Enrichment - Tutor

May. 2023 - May 2025

- Provided one-on-one coaching and group tutoring of up to 30 students in math, engineering, and science courses.

PROJECTS

Advanced Path-Tracking Autonomous Robot

C, MPLAB X IDE, PIC32

- Developed embedded C software for a PIC32-based autonomous robot, integrating infrared sensor inputs, motor control logic, and real-time path-tracking algorithms for autonomous navigation.
- Implemented sensor-based decision-making and actuator response logic under hardware timing constraints.

FireGuard - Innovative Fire Safety and Detection

Python, HTML, Raspberry Pi

- Designed a fire detection system using a Raspberry Pi and thermal cameras, coded in Python.
- Programmed a GUI to monitor temperature in real time and send alerts via text messages

FinShark: Personal Finances App

TypeScript, FastAPI, React

- Developed a full-stack personal finance application enabling users to track expenses and monitor transactions.
- Integrated a FastAPI backend with a TypeScript/React frontend to deliver AI based spending analytics.

Advanced IoT Accelerometer and Temperature Sensor

KiCAD, Arduino Uno, Adafruit IO

- Designed a custom printed circuit board with Arduino, combining motion sensing, temperature tracking, and a display for real-time data monitoring through Adafruit IO.

4-Bit VLSI Arithmetic Logic Unit Design and Implementation

Cadence Design Systems

- Constructed a 4-bit ALU using a hierarchical CMOS design approach and meeting design rule checks.

SKILLS

Languages: C++, C, C#, Python, Javascript, Typescript, MATLAB, html, Verilog, SQL, CSS, Shell, JSON/YAML

Technologies: PyTorch, FastAPI, React, OpenAI, Docker, Git, .NET, WPF, Cadence, Mirage, KiCad, MongoDB, Linux, Arduino, Simulink, AutoCAD, Gemini, Kivy, Docker, MPLAB X IDE,